

ARPG Game Design – Sprint 1

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CITA 412 – Advanced Game Design & Application

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[Link to GitHub](#)

[Link to Jira](#)

Product Backlog

Our product backlog has grown exponentially as we have worked on Sprint 1. We determined that there were many more systems, assets, and logic that need to be added to make our game playable. When we started the project, we had 50 scrums in our product backlog. It has now ballooned to 90.

Part of the reason for our ballooning product backlog is due to our better understanding of the project scope. When we started, we had some confusion about what we were working on. As we progressed through Sprint 1, we began to see the full picture of what our game is going to be.

Here is our current product backlog:

Summary	Assignee	Priority	Status
Enemy 2: Create an Enumerator method that spawns a vertical laser on a cooldown.	Michael	Medium	Done
As a player, I want random events that I can interact with at shops.	Jeremy	Low	To Do
Enemy 2: Create a method that handles enemy hovering movement	Michael	Medium	Done
As a player, I want a shop that I can purchase upgrades in	Jeremy	High	To Do
As a player, I want a map that shows chambers I have explored and where I may go.	Jeremy	Highest	To Do
Enemy 1: Create a method that handles proximity explosion during phase 2	Michael	Medium	To Do
Enemy 1: Create a method that handles enemy death during phase 2.	Michael	Medium	To Do
Enemy 1: Create a method that handles ground movement towards the player in phase 2.	Michael	Medium	To Do

Enemy 1: Create a method that handles enemy transition from phase 1 to phase 2.	Michael	Medium	Done
Create Characters	Jeremy	High	To Do
Create a method that allows the enemy to target the player's location	Michael	Medium	Done
Enemy 1: Create an Enumerator method that picks a direction to move to.	Michael	Medium	Done
As a player, I want to choose chambers to receive its reward	Jeremy	High	To Do
Enemy 1: Create a method that handles movement when hovering in the air.	Michael	Medium	Done
Create a method that handles enemy death.	Michael	Medium	Done
As a dev, I want a script that allows me to generate differently-sized maps	Jeremy	High	Done
Create a method that allows the player to deal damage to the enemy, and vice versa	Michael	Medium	Done
As a player I would like to be able to pause the game at any point.	Michael	Low	To Do
As a player I would like to be able to change the sensitivity of my mouse so I can customize my aiming experience	Michael	Lowest	Done
As a player, I'd like to maneuver a factory.	Aidan	Medium	To Do
Add basic character movement	Jeremy	Highest	Done
As a player, I want to generate random maps	Jeremy	High	Done
Add a functional first person camera	Michael	Highest	Done
Enemy 3: Create a method that moves the enemy in a given direction	Michael	Medium	To Do

Enemy 3: Create a method that handles enemy movement to stay 10ft from the player	Michael	Medium	To Do
Enemy 3: Create a method that spawns projectiles from the enemy	Michael	Medium	To Do
Enemy 3: Create an Enumerator that handles the firing routine	Michael	Medium	To Do
Enemy 2: Create a bool that checks whether or not the enemy should retreat (based on health or time?)	Michael	Medium	To Do
Enemy 2: Create a method that handles enemy death	Michael	Medium	To Do
Enemy 2: Teleports to a pre-baked position in the same chamber as the player	Michael	Medium	To Do
Enemy 2: Create a method that allows the enemy to get the teleport location	Michael	Medium	To Do
Enemy 2: Create a method that allows the enemy to teleport to a location within the chamber it was spawned in.	Michael	Medium	To Do
Enemy 2: Create a method that allows the enemy to knockback the player after taking enough damage.	Michael	Medium	To Do
Add input maps for character input	Jeremy	Highest	Done

Create a method that disables player dashes	Unknown	Medium	To Do
Create a method that slows player movement	Unknown	Medium	To Do
Enemy 5: Create a method that allows the enemy to utilize the dragging method, and grant it a cooldown	Unknown	Medium	To Do
Enemy 5: Create a method that decides whether the enemy should shoot or hook the player	Unknown	Medium	To Do
Enemy 5: Create a method that allows the enemy to shoot at the player	Unknown	Medium	To Do
Enemy 5: Create a chain model that shrinks as the player gets closer to the enemy	Unknown	Medium	To Do

Enemy 5: Create a method that attaches a hook model to the player	Unknown	Medium	To Do
Enemy 5: Create a method that drags the player towards the enemy	Unknown	Medium	To Do
As a player, I want to have the ability to change my key bindings	Jeremy	Low	To Do
Enemy 5: Create a method that allows for slow enemy movement	Unknown	Medium	To Do
Enemy 4: Create a method that allows the enemy to melee the player.	Unknown	Medium	To Do

As a player, I would like to save my run's progress	Jeremy	Highest	To Do
As a Dev, I want layout designs for the chambers to help visualize the chamber placements	Jeremy	Highest	To Do
Create a Shop NPC / Terminal UI screen	Unknown	Medium	To Do
Shop System	Unknown	Medium	To Do
Chamber Design	Unknown	Medium	To Do
Chamber System	Unknown	Medium	To Do

Character 3	Unknown	Medium	To Do
Character 2	Unknown	Medium	To Do
Character 1	Unknown	Medium	To Do
Enemy 5	Unknown	Medium	To Do
Enemy 4	Unknown	Medium	To Do
Enemy 3	Unknown	Medium	To Do
Enemy 2	Unknown	Medium	To Do
Enemy 1	Unknown	Medium	To Do
Add sound controls	Jeremy	High	To Do
As a dev, I want to have an interactable interface for scripts to implement	Jeremy	High	To Do
Create Shop/Chamber Items	Jeremy	Highest	To Do

Implement object pooling for enemy models	Unknown	Medium	To Do
Implement object pooling for projectiles made by enemies	Unknown	Medium	To Do
Create a method that changes player height	Unknown	Medium	To Do

Create Main Menu Music	Unknown	High	To Do
Create SFX for Enemy Interactions	Unknown	Highest	To Do
Create SFX for Player Interactions	Unknown	Highest	To Do
User Interface	Unknown	Medium	To Do
Design game paused UI	Unknown	Medium	To Do
Design controls UI	Unknown	Medium	To Do
Design main menu UI	Unknown	Medium	To Do
As a player, I want a user interface	Unknown	Medium	To Do
As a player, I would like to see my overall statistics for the game	Jeremy	Medium	To Do
As a player, I would like to see my run statistics	Jeremy	High	To Do
Data Collection & Storage	Unknown	Medium	To Do
As a dev, I want a character design document to centralize character abilities, properties, and functionality	Jeremy	Highest	To Do
As a dev, I want scriptable objects to reference character data	Jeremy	High	To Do
As a dev, I want an inherited structure for shared character logic and data	Jeremy	Highest	To Do

Optimization	Unknown	Medium	To Do
Player Settings	Unknown	Medium	To Do
Optimize Map Generation Algorithm(s)	Jeremy	Highest	To Do
Character System	Unknown	Medium	To Do
Player Controls	Unknown	Medium	To Do
As a dev, I want an initialization scene and script to handle singletons and other logic	Jeremy	Highest	To Do
As a player, I want to see a log of my previous runs	Jeremy	High	To Do

Crossbow: Create Projectile Spawner	Michael	Medium	Done
Crossbow: Create Crossbow Projectile Script	Michael	Medium	Done
As a Player, I want props to feel alive (Valve's style of game design)	Jeremy	Medium	To Do
Generate basic room layout for quarry	Aidan	Medium	Done
Generate basic props for quarry	Aidan	Medium	Done
As a player, I want to see custom assets in the chambers I explore	Aidan	Medium	Done
Asset Creation	Unknown	Medium	To Do
Create Hellish theme for map 3	Aidan	Medium	To Do
Create underground chamber theme for map 2	Aidan	Medium	To Do
Create Hallway Assets	Aidan	Medium	To Do
Create chamber Assets	Aidan	Medium	To Do
Implement Sounds into Video Game	Unknown	Highest	To Do
Sound Design	Unknown	Medium	To Do
Create Boss Music	Unknown	Medium	To Do

Create in-game Music for Each Level Theme

Unknown

Low

To Do

Sprint 1 Backlog

Our backlog for sprint 1 is ambitious. Sprint 1 was designed such that the fundamentals for the game were taken care of early. For example, many of the props that will be used in our first map theme were generated, alongside enemy mechanics, a health system, map system, and much more. Here is our Sprint 1 backlog:

Summary	Assignee	Priority	Status
Crossbow: Create Crossbow Projectile Script	Michael	Medium	Done
Crossbow: Create Projectile Spawner	Michael	Medium	Done
Generate basic room layout for quarry	Jeremy	Medium	Done
Generate basic props for quarry	Jeremy	Medium	Done
As a player, I want to see custom assets in the chambers I explore	Jeremy	Medium	Done
As a Dev, I want layout designs for the chambers to help visualize placement	Jeremy	Highest	Done
Add input maps for character input	Jeremy	Highest	Done
Enemy 1: Create a method that handles proximity explosion during phase 2	Michael	Medium	To Do
Enemy 1: Create a method that handles enemy death during phase 2	Michael	Medium	To Do
Enemy 1: Create a method that handles ground movement towards the player in phase 2	Michael	Medium	To Do
Enemy 1: Create a method that handles enemy transition from phase 1 to 2	Michael	Medium	Done
Create a method that allows the enemy to target the player's location	Michael	Medium	Done
Enemy 1: Create an Enumerator method that picks a direction to move to	Michael	Medium	Done
Enemy 1: Create a method that handles movement when hovering in the air	Michael	Medium	Done
Create a method that handles enemy death	Michael	Medium	Done

As a dev, I want a script that allows me to generate differently sized maps	Jeremy	High	Done
Create a method that allows the player to deal damage to the enemy, and vice versa	Michael	Medium	Done
As a player I want to be able to change the sensitivity of my mouse	Michael	Lowest	Done
Add basic character movement	Jeremy	Highest	Done
As a player, I want to generate random maps	Jeremy	High	Done
Add a functional first-person camera	Jeremy	High	Done

Scrum 1

Date: September 16, 2025

Team Member	Completed Since Last Meeting	Planned Before Next Meeting	Blockers
Aiden	- Created scaffolding, wooden pallets, buckets, jersey barrier (concrete), 2 barrels, traffic cone, toolbox, hammer, screwdriver, wooden road block - Continued working on cement mixer	-Planning On Continuing to create Assets for the environment.	None
Jeremy	- Created template for first person controls - Started map generator with randomization seed - Created node system for gamers with randomized points	-Continue logic for chambers and map generation for nodes within chambers	Needs to know connector placement in rooms; door placement must be deliberate
Michael	- Started working on enemy AI and movement	Continue working on Enemy #1	None

- Focused on Enemy #1

Scrum 2

Date: September 18, 2025

Team Member	Completed Since Last Meeting	Planned Before Next Meeting	Blockers
Aiden	- Created crossbow, railgun, rifle, and beam shotgun	- I plan to finish making a room for our game	None
Jeremy	- I have accomplished adding customized chamber structures to the map.	- I plan to finish connecting the chambers and making placeholder chamber designs that the player can move around in. I also plan to finish creating the map algorithm.	None
Michael	- Enemy 1 movement behavior is complete.	I plan to finish adding a second phase to enemies and making a basic weapon in Unity	None

Scrum 3

Date: September 23, 2025

Team Member	Completed Since Last Meeting	Planned Before Next Meeting	Blockers
Aiden	-I have accomplished making shotgun variants for our game. I made one shotgun with a wooden handle, and another with a metal handle. -I exported all of the assets into the Unity project	- I plan to finish making a door for the chamber rooms.	None
Jeremy	- I have accomplished finishing the map generation algorithm for our game and have refactored the code.	- I have accomplished finishing the map generation algorithm for our game and have refactored the code. - I plan to finish the mouse sensitivity settings and more code refactoring	None
Michael	-I have accomplished making a crossbow for our game. -I added two player abilities	I plan to finish adding projectiles for enemy attacks and making two player abilities	None